

SAFETY DATA SHEET

KAYQUAT II

1. Chemical product and company identification

Product name : KAYQUAT II
KAYQUAT II
Product code : 910626
Recommended use and restrictions : Sanitizer.
Manufacturer/Supplier Address/Telephone : Kay Chemical Company c/o Ecolab G.K.
1-31-18, Takadanobaba, Shinjuku-ku
Tokyo 169-8631, Japan
0120-103-932 (Toll Free)
Emergency telephone number : 0120 - 756 - 005 (Toll Free)

2. Hazards identification

GHS Classification : ACUTE TOXICITY: ORAL - Category 4
ACUTE TOXICITY: SKIN - Category 4
SKIN CORROSION/IRRITATION - Category 1B
SERIOUS EYE DAMAGE/ EYE IRRITATION - Category 1
AQUATIC TOXICITY (ACUTE) - Category 2

GHS label elements

Symbol :



Signal word : Danger

Hazard statements : Harmful if swallowed or in contact with skin.
Causes severe skin burns and eye damage.
Toxic to aquatic life.

Precautionary statements

Prevention : Wear protective gloves. Wear eye or face protection. Wear protective clothing.
Avoid release to the environment. Do not eat, drink or smoke when using this product. Wash hands thoroughly after handling.

Response : IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER or physician. IF SWALLOWED: Immediately call a POISON CENTER or physician. Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. Wash contaminated clothing before reuse. Immediately call a POISON CENTER or physician. IF ON SKIN: Wash with plenty of soap and water. Call a POISON CENTER or physician if you feel unwell. Take off contaminated clothing and wash it before reuse. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER or physician.

Storage : No other specific measures identified.

Disposal : Dispose of contents and container in accordance with all local, regional, national and international regulations.

Other hazards : None known.

3. Composition/information on ingredients

Nature of material : Mixture

Hazardous ingredients	Concentration Range (%)	CAS number
ethanol	< 10	64-17-5
DICAPRYL/DICAPRYLYL DIMONIUM CHLORIDE	Trade secret	Trade secret
cationic surfactant	Trade secret	Trade secret

4. First aid measures

First aid measures

- Eye contact** : Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.
- Skin contact** : Take off immediately all contaminated clothing. Rinse skin with water or shower. Get medical attention immediately. Wash clothing before reuse. Clean shoes thoroughly before reuse.
- Inhalation** : Remove victim to fresh air and keep at rest in a position comfortable for breathing. Get medical attention immediately.
- Ingestion** : Get medical attention immediately. Rinse mouth. Do not induce vomiting.

Most important symptoms/effects, acute and delayed

- Eye contact** : Causes serious eye damage.
- Skin contact** : Causes severe burns. Harmful in contact with skin.
- Inhalation** : May cause respiratory irritation.
- Ingestion** : Harmful if swallowed. May cause burns to mouth, throat and stomach.

- Protection of first-aiders** : No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation. Wash contaminated clothing thoroughly with water before removing it, or wear gloves.
- Notes to physician** : Treat symptomatically. Contact poison treatment specialist immediately if large quantities have been ingested or inhaled.

See toxicological information (Section 11)

5. Fire-fighting measures

- Suitable fire extinguishing media** : Use water spray, fog or foam.
- Specific hazards arising from the chemical** : In a fire or if heated, a pressure increase will occur and the container may burst. This material is toxic to aquatic life. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain.
- Hazardous thermal decomposition products** : Decomposition products may include the following materials:
carbon dioxide
carbon monoxide
- Specific fire-fighting methods** : Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. No action shall be taken involving any personal risk or without suitable training.
- Special protective equipment for fire-fighters** : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

6. Accidental release measures

- Personal precautions** : Initiate company's spill response procedures immediately. Keep people out of area. Put on appropriate personal protective equipment (see section 8). Do not touch or walk through spilled material.
- Environmental precautions** : Avoid contact of spilled material and runoff with soil and surface waterways.
- Methods for cleaning up** : Follow company's spill procedures. Keep people away from spill. Put on appropriate personal protective equipment (see section 8). Absorb/neutralize liquid material. Use a tool to scoop up solid or absorbed material and put into appropriate labeled container. Use a tool to scoop up solid or absorbed material and place into appropriate labeled waste container. Use a water rinse for final clean-up.

7. Handling and storage

- Handling** : Do not ingest. Do not get in eyes or on skin or clothing. Do not breathe vapor or mist. Use only with adequate ventilation. Wash thoroughly after handling.
- Storage** : Keep out of reach of children. Keep container tightly closed.

Store between the following temperatures: 0 and 50°C

8. Exposure controls/personal protection

Control parameters

Ingredient name	Exposure limits
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- Appropriate engineering controls** : Use only with adequate ventilation. If user operations generate dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Personal protection

- Hygiene measures** : Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Appropriate techniques should be used to remove potentially contaminated clothing. Wash contaminated clothing before reusing.
- Eye protection** : Use chemical splash goggles. For continued or severe exposure wear a face shield over the goggles.
- Hand protection** : Use chemical-resistant, impervious gloves.
- Skin protection** : Use synthetic apron, other protective equipment as necessary to prevent skin contact.
- Respiratory protection** : A respirator is not needed under normal and intended conditions of product use.

9. Physical and chemical properties

Appearance

- Physical state** : Liquid.
- Color** : Red
- Odor** : Faint odor
- pH** : 6 to 9 (100%)
- Melting point** : Not available.
- Boiling point** : >100°C (>212°F)
- Flash point** : > 100°C
- Evaporation rate (butyl acetate = 1)** : Not available.

9. Physical and chemical properties

Flammability (solid, gas)	: Not available.
Explosion limits	: Not available.
Vapor pressure	: Not available.
Vapor density	: Not available.
Relative density	: 0.984 to 1.008 (Water = 1)
Solubility	: Easily soluble in the following materials: cold water and hot water.
Partition coefficient: n-octanol/water	: Not available.
Auto-ignition temperature	: Not available.

10. Stability and reactivity

Stability	: The product is stable.
Possibility of hazardous reactions	: Under normal conditions of storage and use, hazardous reactions will not occur.
Conditions to avoid	: No specific data.
Materials to avoid	: Not available.
Hazardous decomposition products	: Under normal conditions of storage and use, hazardous decomposition products should not be produced.

11. Toxicological information

Symptoms

Eye contact	: Adverse symptoms may include the following: pain watering redness
Skin contact	: Adverse symptoms may include the following: pain or irritation redness blistering may occur
Inhalation	: Adverse symptoms may include the following: coughing Respiratory tract irritation
Ingestion	: Adverse symptoms may include the following: stomach pains

Acute toxicity

Eye contact	: Causes serious eye damage.
Skin contact	: Causes severe burns. Harmful in contact with skin.
Inhalation	: May cause respiratory irritation.
Ingestion	: Harmful if swallowed. May cause burns to mouth, throat and stomach.

Toxicity data

Product/ingredient name	Result	Species	Dose	Exposure
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11. Toxicological information

quaternary ammonium compounds, di-c8-10-alkyldimethyl, chlorides	LD50 Dermal	Rat	1500 mg/kg	-
	LD50 Oral	Rat	366 mg/kg	-
cationic surfactant	LD50 Dermal	Rabbit	3340 mg/kg	-
	LD50 Oral	Rat	344 mg/kg	-
ethanol	LD50 Dermal	Rabbit	15800 mg/kg	-
	LD50 Oral	Rat	10470 mg/kg	-
	LC50 Inhalation Vapor	Rat	117 mg/l	4 hours

Chronic toxicity

Carcinogenicity	: No known significant effects or critical hazards.
Mutagenicity	: No known significant effects or critical hazards.
Teratogenicity	: No known significant effects or critical hazards.
Developmental effects	: No known significant effects or critical hazards.
Fertility effects	: No known significant effects or critical hazards.

12. Ecological information

Ecotoxicity : This material is toxic to aquatic life.

Aquatic and terrestrial toxicity

Product/ingredient name	Test	Result	Species	Exposure
quaternary ammonium compounds, di-c8-10-alkyldimethyl, chlorides	-	Acute LC50 <1 mg/l	Fish	96 hours
cationic surfactant	-	Acute LC50 0.515 mg/l	Fish	96 hours
ethanol	-	Acute LC50 11000 mg/l	Fish	96 hours

Persistence and degradability

Product/ingredient name	Test	Result	Dose	Inoculum
Not available.				

<u>Product/ingredient name</u>	<u>Aquatic half-life</u>	<u>Photolysis</u>	<u>Biodegradability</u>
Not available.			

Bioaccumulative potential

<u>Product/ingredient name</u>	<u>LogP_{ow}</u>	<u>BCF</u>	<u>Potential</u>
ethanol	-0.31	-	low

Mobility in soil : Not available.

Other adverse effects : No known significant effects or critical hazards.

13. Disposal considerations

Disposal methods : The generation of waste should be avoided or minimized wherever possible. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements. Dispose of surplus and non-recyclable products via a licensed waste disposal contractor. Waste should not be disposed of untreated to the sewer unless fully compliant with the requirements of all authorities with jurisdiction. Waste packaging should be recycled. Incineration or landfill should only be considered when recycling is not feasible. This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers.

14. Transport information

Certain shipping modes or package sizes may have exceptions from the transport regulations. The classification provided may not reflect those exceptions and may not apply to all shipping modes or package sizes.

UN number : UN3082

Proper shipping name : ENVIRONMENTALLY HAZARDOUS
SUBSTANCE, LIQUID, N.O.S.
(Quaternary ammonium compounds)
Marine pollutant

Class : 9

Packing group : III

Special Safety Measures and Conditions: During transport, avoid direct sunlight, container damage, corrosion, loading to ensure that all work will be done to ensure the prevention of collapse of cargo.

15. Regulatory information

Law Concerning : Marine pollutant: P
Prevention of Pollution of
the Ocean and Maritime
Disaster

	<u>Chemical name</u>	<u>Class</u>	<u>Number</u>	<u>%</u>
Pollutant Release and Transfer Registers (PRTR)	: Not applicable.			

ISHL Chemicals requiring notification	: Ethanol	% < 10
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ISHL Dangerous : Inflammable
Substances

Safety, health and : No known specific national and/or regional regulations applicable to this product
environmental regulations (including its ingredients).
specific for the product

16. Other information

References

- 1) Public relations companies and pharmacy standards poisonous substances notification Album
- 2) RTECS
- 3) International Technical Information Institute manual for handling dangerous poison
- 4) 12,996 chemical products "Chemical Daily"
- 5) Data safety inspections of existing chemical substances Law (Chemicals Inspection Association)
- 6) Daily Chemical Industry Chemical Safety Data Book
- 7) All chemical substances subject to daily data for Occupational Safety and Health Act
- 8) Japan Information Center for chemical safety
- 9) GHS Purple Book Ver. 2

History

Date of issue : 2013 . 3 . 26

Company Name : ECOLAB

Department : Regulatory Affairs

Notice to reader

16. Other information

The above information is believed to be correct with respect to the formula used to manufacture the product in the country of origin. As data, standards, and regulations change, and conditions of use and handling are beyond our control, NO WARRANTY, EXPRESS OR IMPLIED, IS MADE AS TO THE COMPLETENESS OR CONTINUING ACCURACY OF THIS INFORMATION.